



— EXECUTIVE PROGRESS DOSSIER

# ICPC HUE Community Progress Report

From Winter 2025 Launch to National ECPC Qualification and the 2026/2027 Operating Model

## Executive Retrospective

Since its launch on November 28, 2025, ICPC HUE transformed from a student initiative into an institutional competitive programming powerhouse at Horus University. Over the 2025/2026 season, the community engineered a custom digital platform serving thousands of monthly visitors, delivered 17+ hours of live curriculum, hosted a landmark Orientation Day with national champions, crowned internal champions, and qualified seven fully-formed collegiate teams for ECPC. This report synthesizes our foundational achievements, the DCC and ECPC milestones, and the roadmap for 2026/2027.



ECPC Qualification Delegation

*Horus University Competitive Programming Delegation at the ECPC Qualification Stage — Arab Academy for Science, Technology and Maritime Transport (AASTMT), August 2026.*



## — INSTITUTIONAL TRAJECTORY &amp; ROADMAP

# The Road to DCC: Building Our Competitive Foundation

A strategic overview of how ICPC HUE established a collegiate training ecosystem, engineered a custom learning platform, and formed seven national teams before our regional debut.

**650****CURATED PROBLEMS**

Structured across 3 skill levels (Levels 0–2) on the custom [icpchue.com](https://icpchue.com) platform.

**17+ hrs****LIVE TRAINING**

Structured live lectures delivered covering foundational syntax to algorithmic logic.

**120****ACTIVE USERS**

Active student users on [icpchue.com](https://icpchue.com) engaging with problem sheets and submissions.

**16 Teams****HORUS DELEGATION**

16 teams from all Horus University, including 7 teams from Faculty of AI directly trained by ICPC HUE.

**01****Community Inception & Winter Camp Launch**

DEC 2025 – JAN 2026

Established the competitive programming initiative at Horus University; launched a university-wide Winter Camp welcoming beginners starting from scratch while activating senior students.

**02****The ICPC HUE Educational Platform ([icpchue.com](https://icpchue.com))**

FEBRUARY 2026

Engineered and deployed our custom training hub with official university email authentication, automated problem evaluation, and categorized sheets from Level 0 to Level 2.

**03****The Landmark Orientation Day (Faculty of AI)**

MARCH 12, 2026

Under the sponsorship of Dr. **Basma Mostafa** & Dr. **Yasser El-Awady** and supervision of **Eng. Fatima Magdy** & **Eng. Raafat Mohamed**, hosted hundreds of students and six decorated ACPC DU national champions to inspire our community.

**04****High-Stakes Selection Trials & Campus Champions**

APRIL 2026

Conducted two official selection contests to evaluate algorithmic mastery and team synergy, crowning top campus solvers across competitive individual trials and finalizing collegiate lineups.

**05****Seven AI Teams Formed & Approved for National ECPC**

APRIL 30 – MAY 2026

Assembled 21 student competitors into 7 balanced three-member teams from the Faculty of AI (4 Level 1 teams, 3 Level 2 teams), completing official registration for the Egyptian Collegiate Programming Contest as part of the university delegation.

**Strategic Proving Ground:** With seven official teams assembled and our digital training platform established, ICPC HUE partnered with ACPC Club Damietta University to co-organize DCC 2026 as an authentic regional tournament simulation.

## REGIONAL PROVING GROUND

# DCC 2026: Competitive Programming in Damietta

Co-organized with ACPC Club Damietta University as a practical pre-ECPC simulation and regional milestone.

### 2 Phases

ONLINE & IN-PERSON FINAL

### 5 Communities

MNU, HUE, DELTA, DU, NDETI

### 100+ Contestants

FIRST DAMIETTA EDITION

#### DCC Presentation



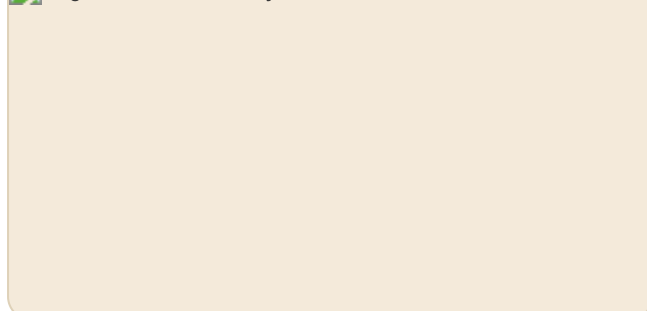
Co-founder of ICPC HUE speaking at DCC.

#### First Place Award Ceremony



First Place winners receiving their championship check.

#### Organizers and Faculty with Students



Faculty leadership, community organizers, and student competitors at DCC.

#### Group Photo of Contestants



Group photo of all the 100 contestants of DCC.

### Digital Infrastructure: dcchub.xyz

ICPC HUE member **Yousef Mohamed** developed **dcchub.xyz**, serving as the official digital portal for registration, announcements, and live contest standings.

### Faculty Supervision & Support

Delivered under Dean **Dr. Hossam El-Din Mostafa**, coordinators **Dr. Basma Mostafa & Dr. Yasser El-Awady**, and supervising engineers **Fatma Magdy, Sara Lotfy, and Raafat El-Moneer**.

## NATIONAL FIELD EXPERIENCE

# ECPC Qualification: Turning Preparation into Evidence

A dedicated one-month intensive curriculum culminated in delegation travel to AASTMT and competitive validation.

### The Tournament Journey

Approximately one month prior to ECPC qualification, ICPC HUE conducted rigorous reviews of advanced data structures, completed official collegiate registration, and organized travel logistics.

#### AUGUST 18 DEPARTURE

Delegation travelled to Alexandria, checked into accommodations, and participated in community team-building.

#### AASTMT CAMPUS TESTING

Teams completed the official practice contest, verified IDE/PC<sup>2</sup> setups, received team jerseys, and acclimatized to tournament pressure.

#### OFFICIAL COMPETITION ROUND

Competed against premier universities nationwide over 5 rigorous hours, demonstrating tactical problem triage and high-pressure execution.

## 12

#### PROBLEMS

Challenging official problem set.

## 3+

#### SOLVED MOST

Baseline reached by majority.

## 4

#### TOP SCORE

Within 1 problem of advancing.



Team at ECPC Qualification

Horus University team working through algorithmic problem sets during the official qualification round.

### Key Takeaways for Next Season: Within 1 Problem of National Qualification

Horus University proved its competitive caliber by finishing within **just 1 single problem** of advancing directly to the ECPC National Finals. The results strongly validated student capability while pinpointing precise areas for growth: starting structured multi-hour practice contests months earlier, instituting daily mentor follow-ups, and conducting timed team selection trials.

## OUTREACH & COMMUNITY MOMENTUM

# Converting Contest Experience into Season-Wide Momentum

Documenting the tournament journey transformed competitive programming into an exciting, high-visibility activity across Horus University.

### 43K+ Facebook Views

REACH

A synchronized campaign published travel footage and updates, generating **43,000+ views on Facebook** and establishing competitive programming as a recognized university endeavor.

### Educational Video Series

OUTREACH

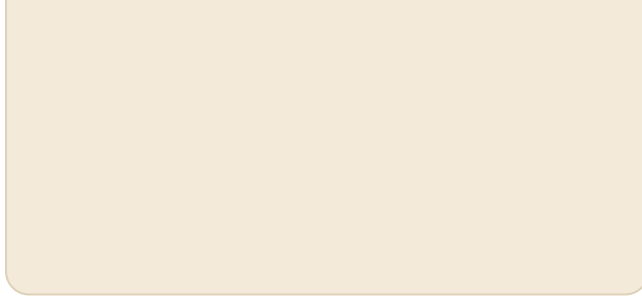
Three professionally edited videos welcomed incoming freshmen, illustrating problem-solving concepts, showcasing student achievements, and outlining upcoming training cohorts.

### Hire Up Recruitment

INTAKE

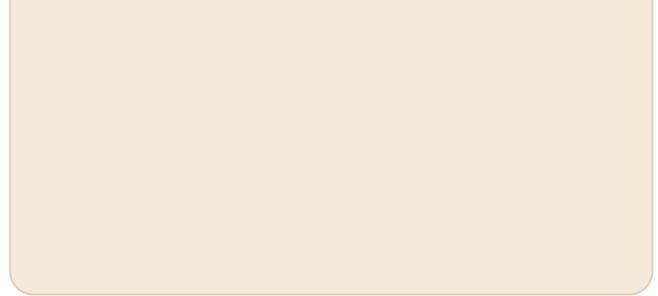
Launched recruitment via [icpchue.com/job](https://icpchue.com/job) for student instructors, mentors, media creators, and operations officers to formalize staffing.

### Delegation Bus Travel to Alexandria



*Delegation transit to Alexandria: building community synergy, morale, and focus.*

### Delegation Tournament Celebration



*Post-contest tournament celebration on the AASTMT Alexandria campus lawn.*

### Institutional Impact & Recruitment Surge

The high-visibility media coverage and viral social media reception directly accelerated community recruitment for the 2026/2027 season. Over 300 prospective freshmen and returning students registered interest via [icpchue.com/job](https://icpchue.com/job) within the first 72 hours, laying the groundwork for our largest and most competitive intake to date.

## NATIONAL COLLEGIATE DELEGATION

# Our 16 Competing Teams at ECPC Qualifications

48 student competitors representing the Faculty of Artificial Intelligence and Faculty of Engineering across 16 official tournament teams at AASTMT, Alexandria.

### 16 Teams

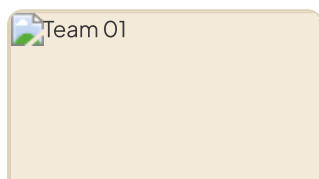
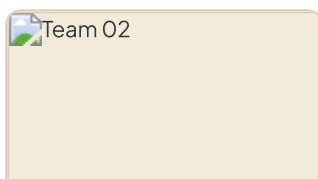
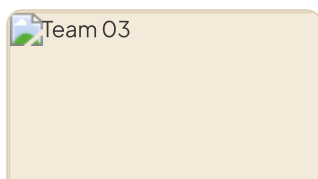
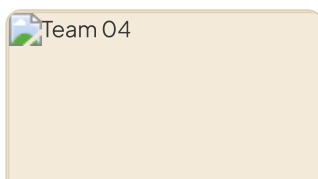
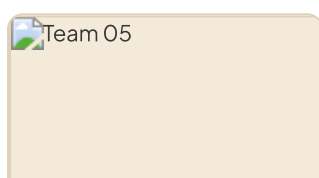
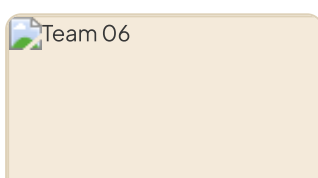
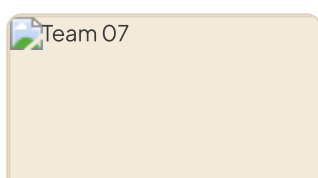
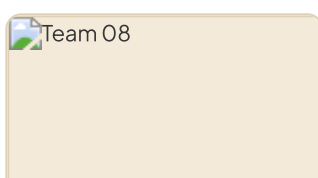
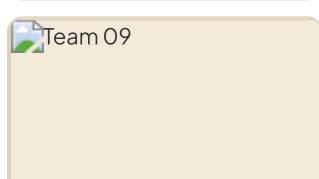
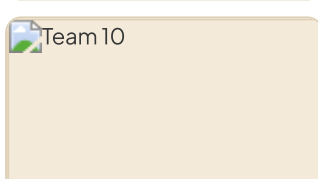
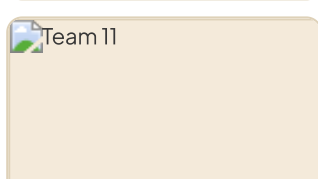
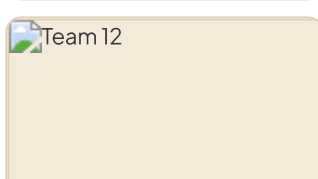
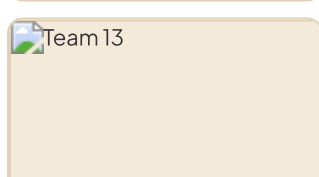
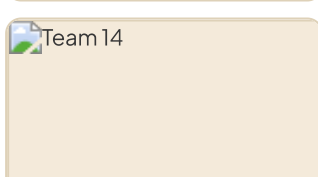
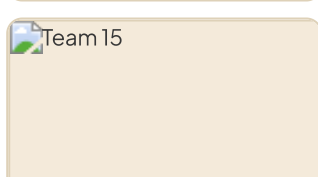
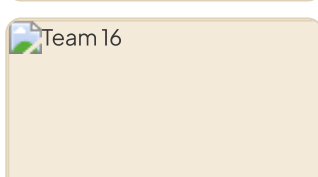
OFFICIAL DELEGATION

### 48 Students

ACTIVE COMPETITORS

### 2 Faculties

AI &amp; ENGINEERING

|                                                                                                |                                                                                                |                                                                                                 |                                                                                                  |
|------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| <br>Team 01   | <br>Team 02   | <br>Team 03   | <br>Team 04   |
| <br>Team 05   | <br>Team 06   | <br>Team 07   | <br>Team 08   |
| <br>Team 09 | <br>Team 10 | <br>Team 11 | <br>Team 12 |
| <br>Team 13 | <br>Team 14 | <br>Team 15 | <br>Team 16 |

## — OPERATING MODEL

# The Four Community Committees: Structured Specialization

A defined functional division of labor designed to overcome technical gaps, inconsistent follow-up, and logistical overhead.

**Foundational Leadership Principle:** No member is forced into a particular committee. Each student actively selects the role matching their skills and passion, ensuring high intrinsic accountability and pedagogical integrity.



## 01 · Instructors Committee

Deliver weekly technical lectures, solve problems live with step-by-step clarity, and prepare structured problem sheets. Instructors must demonstrate academic honesty, deep algorithmic mastery, and the patience required to guide novice programmers.



## 02 · Mentors Committee

Track small teams continuously, conduct weekly check-ins, identify when students encounter conceptual roadblocks, and monitor student wellbeing. Mentors ensure students maintain consistent practice habits rather than dropping off.



## 03 · Media Committee

Produce high-quality presentations, promotional videos (using Adobe Premiere and After Effects), and social media documentation. They celebrate student achievements, elevate community visibility, and drive recruitment.



## 04 · Operations Committee

Manage physical logistics, coordinate lab reservations with university administration, organize on-campus event flows, oversee printing of materials, and ensure smooth operational execution during all offline contests.

## TALENT PIPELINE & SYSTEMS

# Leadership Development and the ICPC HUE Platform

Identifying student talent through rigorous teach-backs and unifying mentoring workflows into a closed-loop platform.

## Top-5 Team Leadership Challenge

To prepare the next generation of student instructors, ICPC HUE conducted extensive interviews with the **top five teams** from the ECPC qualification stage: four teams from the Faculty of Artificial Intelligence and one from Engineering.

### The 5-Day "Teach-Back" Evaluation

Candidates were assigned complex technical subjects (STL vectors, iterators, and recursion) to study over 5 days. They then conducted live "teach-back" presentations evaluated for pedagogical clarity, calmness under questioning, and ability to inspire peers.

Interviews and evaluations were conducted with the guidance of faculty engineers: **Eng. Fatma Magdy, Eng. Sara Lotfy, and Eng. Raafat El-Moneer.**

## The ICPC HUE Platform: Closed-Loop Tracking

The digital platform was developed to serve as the central backbone for all community recruitment, problem submission, and mentor reviews.

### 1. Daily Student Problem Logs

Students record sheets and solved problems daily, encouraging self-reflection and habit formation.

### 2. Real-Time Mentor Dashboard

Mentors view active solving volume, spotting participation drops before students fall permanently behind.

### 3. Targeted Technical Interventions

Instructors tailor weekly review sessions based on actual aggregate failure rates on specific sheets.

**Result: Transition from informal follow-up to a verifiable training system.**

**Institutional Outcome:** By connecting student logs, mentor monitoring, and instructor reviews into a single system, ICPC HUE ensures that student progress is quantitatively tracked throughout the academic year.

## ECOSYSTEM &amp; CAMPUS ENGAGEMENT

# Strategic Partnerships & Community Introduction Day

Expanding Horus University's learning network and preparing a professional orientation for incoming freshmen.

## Strategic Regional Partnerships

3 Partner Communities

## KNOWLEDGE EXCHANGE & CO-HOSTING



### ACPC Club Damietta Univ.

Established through the joint organization of DCC 2026. This ongoing relationship enables shared problem setting, contest hosting infrastructure, and combined local competitions.



### ICPC PSU Community Port Said University

A direct connection to a seasoned regional community with proven ECPC bronze-medal experience, creating valuable peer mentoring opportunities for advanced HUE teams.



### ICPC Assiut Community Assiut University

Cooperation focused on pedagogical standards and instructor training, drawing upon one of the largest and most successful competitive programming ecosystems in Egypt.

## Community Introduction Day Preparations

Freshmen Intake 2026/27

To welcome incoming students and establish professional standards from day one, ICPC HUE developed a comprehensive set of event assets:



### Award Certificates

Honoring top 5 ECPC teams



### Financial Prizes

Reward checks for top achievers



### Official T-Shirts

Branded community apparel



### Printed Flyers

Season roadmap & intake guides



### Flag & Banners

Official HUE roll-ups and flags



### Welcome Gifts

Chocolates and intake merchandise



### Keynote Speaker: Official Collegiate Guest Address

Contacted an ICPC-qualified veteran to deliver a guest address on real-world problem-solving value, algorithmic careers, and the discipline needed to excel in collegiate contests.



## ANNUAL ROADMAP

# The 2026/2027 Road to ICPC: Season Pathway

A 7-phase algorithmic curriculum guiding students from basic programming fundamentals to national ECPC selection.



## Road to ICPC Pathway

| Phase & Stage                       | Timeline                    | Core Technical Focus & Milestone Events                                                       |
|-------------------------------------|-----------------------------|-----------------------------------------------------------------------------------------------|
| Phase 1: Recruitment & Orientation  | Aug 29 – Oct 9, 2026        | Campus promotion, online forms, interviews, Orientation Day, Intro Session.                   |
| Phase 2: Programming Foundations    | Oct 16 – Nov 26, 2026       | Variables, loops, arrays, strings, functions, Big-O complexity; weekly practice contests.     |
| Phase 3: STL & Winter Checkpoint    | Nov 27, 2026 – Jan 29, 2027 | Vectors, stacks, queues, sets, maps, priority queues;<br><b>Warm-up Contest: Jan 29, 2027</b> |
| Phase 4: Core Problem-Solving       | Feb 5 – Mar 19, 2027        | Frequency arrays, two pointers, binary search, greedy algorithms, recursion, backtracking.    |
| Phase 5: Advanced Algorithms        | Mar 26 – Apr 30, 2027       | Mathematics I & II, Dynamic Programming (DP I & II), Graph Algorithms I.                      |
| Phase 6: Team Formation & Selection | Apr 24 – Jun 18, 2027       | Internal team trials, Team Selection Contest (TSC), official team roster announcement.        |
| Phase 7: Final ECPC Preparation     | Jun 22 – Aug 15, 2027       | Intensive weekly 5-hour simulated contests, full tournament simulation, target ECPC 2027.     |

\* Unless otherwise stated, technical sessions are held Fridays at 8:30 PM. Dates marked as estimated adjust in accordance with the official Horus University academic calendar.

## ACADEMIC APPENDIX & MILESTONES

# 2026/2027 Season Curriculum & Milestones

Comprehensive schedule of all 46 milestones across the academic year for faculty review and operational tracking.

### Term 1: Foundations & STL

01. Promotion: Aug 29 – Sep 03
02. Applications Open: Sep 05 – Sep 10
03. Screening/Interviews: Sep 12 – Sep 17
04. Results: Sep 18
05. Orientation Day: Sep 26 – Oct 01
06. Form Reopen: Sep 26 – Oct 01
07. Intro Session: Oct 09 (8:30 PM)
08. Basics & Conditions: Oct 16
09. Loops & Nested Loops: Oct 23
10. Arrays & Functions: Oct 30
11. Strings: Nov 06
12. Midterm Break: Nov 07 – Nov 12
13. Functions: Nov 13
14. Complexity (Big-O): Nov 20
15. Practice Contests: Nov 21 – Nov 26
16. Progress Check: Late Nov
17. STL Vector & Pair: Nov 27
18. STL Stack & Queue: Dec 04
19. STL Set & Map: Dec 11
20. STL Priority Queue: Dec 18
21. Finals Break: Dec 20 – Jan 21
22. **Warm-up Contest: Jan 29, 2027**

### Term 2: Core Algorithms

23. Prefix & Freq Arrays: Feb 05
24. Two Pointers / Sliding Win: Feb 12
25. Form Reopen: Feb 13 – Feb 18
26. Binary Search: Feb 19
27. Greedy Algorithms: Feb 26
28. Recursion & Brute Force: Mar 05
29. Diagnostic Review: Mar 06 – Mar 11
30. Backtracking: Mar 12
31. Bitmask & Brute Force: Mar 19
32. Mathematics I: Mar 26
33. Mathematics II: Apr 02
34. Dynamic Programming I: Apr 09
35. Progress Check: Apr 10 – Apr 15
36. Midterm Break: Apr 17 – Apr 22
37. Dynamic Programming II: Apr 23
38. Graph Algorithms I: Apr 30

### Summer: Selection & ECPC

39. Team Trials: Apr 24 – May 06
40. Team Selection Contest: May 08
41. Official Teams Announced: May 15
42. Eid Break: May 16 – May 19
43. Finals Break: May 22 – Jun 18
44. Weekly Contests: Jun 22 – Jul 27
45. Full Contest Simulation: Aug 01
46. **Target ECPC 2027: Aug 15**

#### Governance Note:

All technical content is peer-reviewed prior to delivery. Attendance, platform problem submissions, and contest logs are monitored continuously.

## STRATEGIC VISION & LONG-TERM TRAJECTORY

# Conclusion and Next Strategic Objectives

The full 2025/2026 trajectory proved that ICPC HUE can organize, compete, learn, and build a lasting system simultaneously.

### 1 Expand Instructor & Mentor Pipeline

Institutionalize the teach-back interview framework to groom past top performers into certified student instructors and mentors, preventing knowledge loss as senior cohorts graduate.

### 2 Standardize Daily Platform Tracking

Make daily submission logging on the ICPC HUE platform mandatory for all active trainees, providing mentors with transparent data to intervene early when solving momentum dips.

### 3 Deepen Cross-University Cooperation

Expand joint contests and instructor exchanges with Assiut University, Port Said University, and Damietta University, ensuring HUE students are regularly benchmarked against top regional teams.

### 4 Initiate Structured Simulation Earlier

Transition teams into 5-hour timed contest simulations months ahead of ECPC qualification, building the endurance, strategic problem triage, and team dynamics required to advance.

### Institutional Long-Term Objective

Establish a self-sustaining competitive programming culture at Horus University — guiding students from their very first solved problem to the highest levels of collegiate excellence in the ECPC, ACPC, and ICPC World Finals.

*"ICPC HUE is continuing to build that path one student, one team, and one well-organized training cycle at a time."*



# Meet the Team

## THE PEOPLE BEHIND ICPC HUE

ICPCHUE's success is built on the dedication and expertise of our talented team members. From visionary leadership to experienced instructors, each member plays a crucial role in shaping the future of competitive programming at Horus University.

### Academic Supervision & Support

- Dr. Basma Mostafa
- Dr. Yasser Elawady
- Eng. Fatma Magdy
- Eng. Raafat Mohamed
- Eng. Sara

### Leadership

- AbdelRahman Mohsen El-Shahat – Owner
- Youssef Mohamed Salah El-Din – Co-Founder
- Ahmed Waleed – The Lead

### Platform Developing

- Youssef Mohamed Salah El-Din – Main Dev

### Instructors

- Ahmed Waleed – Lead Instructor
- Ibrahim Matar – Co-Lead Instructor
- Omar El-Shayyal – Instructor

### Monitors

- Noha – Lead Monitor

## — INSTITUTIONAL SUPPORT &amp; GOVERNANCE

## Academic Supervision & Special Thanks

With profound gratitude to university leadership, faculty deanship, and supervising engineers who championed our community every step of the way.

**Prof. Dr. Hossam El-Din Mostafa**

DEAN OF FACULTY OF ARTIFICIAL INTELLIGENCE

We express our deepest gratitude to the Dean for his visionary leadership, continuous patronage of collegiate competitive programming, endorsement of university computer laboratory allocations, and championing our participation in regional and national arenas.

**Dr. Basma Mostafa & Dr. Yasser El-Awady**

ACADEMIC PROGRAM COORDINATORS

Heartfelt thanks for their generous sponsorship of the landmark March 12 Orientation Day, providing continuous strategic guidance, bridging faculty resources with student initiatives, and mentoring teams throughout the entire competitive season.

**Eng. Fatma Magdy • Eng. Sara Lotfy • Eng. Raafat Mohamed**

FACULTY SUPERVISING ENGINEERS

Our sincere appreciation for their tireless hands-on mentorship, conducting rigorous 5-day teach-back evaluations for student instructors, coordinating logistics across departments, and providing steadfast academic and moral guidance to every competitor.

**INSTITUTIONAL TRIBUTE TO HORUS UNIVERSITY LEADERSHIP**

*"On behalf of all ICPC HUE competitors, instructors, and student members, we express our highest appreciation to the University Leadership, the Faculty Deanship, and our supervising engineers. Your unwavering trust and academic guidance transformed a student initiative into an official collegiate powerhouse. We pledge to continue building on this foundation and carrying Horus University's flag to the highest podiums in the ECPC, ACPC, and ICPC World Finals."*